

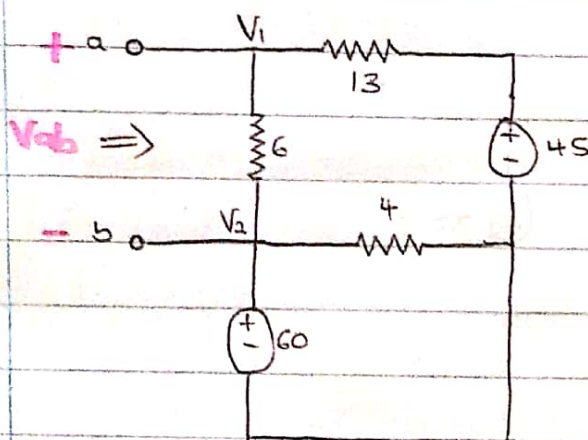
20 September 2022

* equivalent thevenin means R_{th} and V_{th}

Recap

- R_{th} - close all voltage sources
- remove current sources
- don't touch dependent sources

Example 1



* Calculate V_{th}

> you need to find V_1 and V_2

$$V_2 = 60$$

$$\therefore V_1 - V_2 = V_{ab} = V_{th}$$

$$\frac{V_1 - 60}{6} + \frac{V_1 - 45}{13} = 0$$

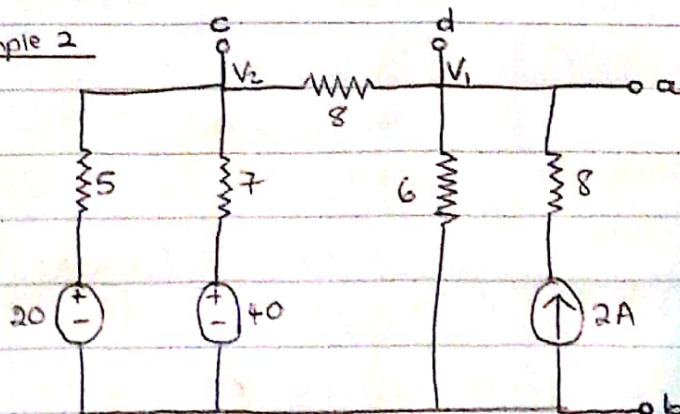
$$\frac{V_1}{6} - \frac{60}{6} + \frac{V_1}{13} - \frac{45}{13} = 0$$

$$\frac{V_1}{6} + \frac{V_1}{13} = \frac{60}{6} + \frac{45}{13}$$

$$\frac{19}{78} V_1 = \frac{175}{13}$$

$$V_1 = 55.26 \text{ V}$$

Example 2



Calculate V_{th} with respect to a-b

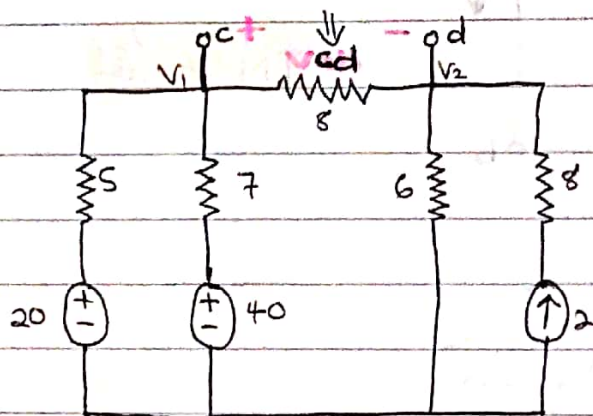
$$\text{@ } V_1: \frac{V_1 - V_2}{8} + \frac{V_1}{6} - 2 = 0 \quad \text{@ } V_2: \frac{V_2 - V_1}{8} + \frac{V_2 - 40}{7} + \frac{V_2 - 20}{5} = 0$$

$$\frac{V_1}{8} - \frac{V_2}{8} + \frac{V_1}{6} = 2 \quad \frac{V_2}{8} + \frac{V_2}{7} + \frac{V_2}{5} - \frac{V_1}{8} = \frac{40}{7} + \frac{20}{5}$$

$$\frac{V_1}{8} + \frac{V_1}{6} - \frac{V_2}{8} = 2 \quad \frac{131}{280} V_2 - \frac{1}{8} V_1 = \frac{68}{7}$$

$$V_1 = 17.79 \quad V_2 = 25.52$$

$$\therefore V_{th} = V_1$$



V_{th} seen by c-d
* calculate the voltage of the 8Ω

$$\text{@ } V_1: \frac{V_1 - 20}{5} + \frac{V_1 - 40}{7} + \frac{V_1 - V_2}{8} = 0 \quad \text{@ } V_2: \frac{V_2 - V_1}{8} + \frac{V_2}{6} - 2 = 0$$

$$\frac{V_1}{5} - \frac{20}{5} + \frac{V_1}{7} - \frac{40}{7} + \frac{V_1}{8} - \frac{V_2}{8} = 0 \quad \frac{V_2}{8} + \frac{V_2}{6} - \frac{V_1}{8} = 2$$

$$\frac{131}{280} V_1 - \frac{V_2}{8} = \frac{68}{7}$$

$$V_1 = 25.52 \text{ V} \quad V_2 = 17.79 \text{ V}$$

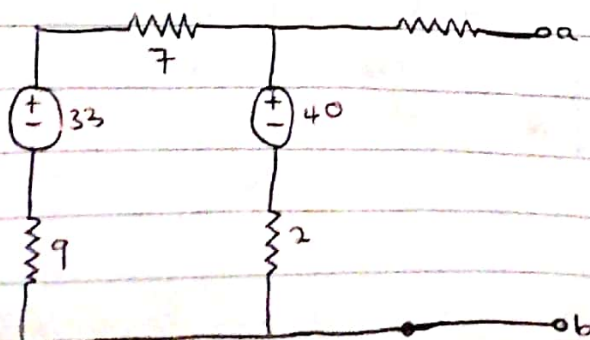
$$V_1 - V_2 = V_{cd}$$

NORTON

$$R_N = R_{th} = R_L \text{ (R-load)}$$

* we are not going to do R_N because it's the same thing as R_{th}

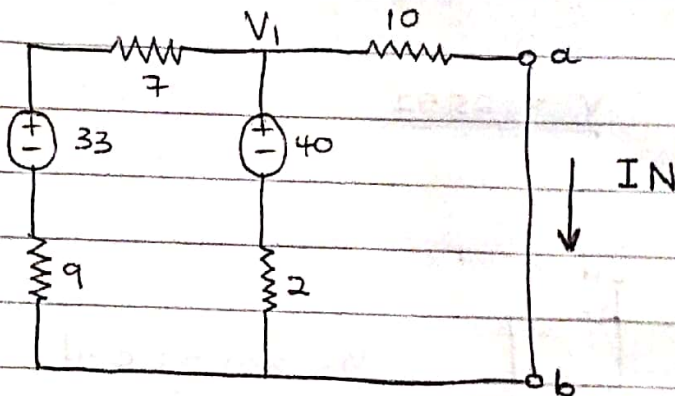
Example 1



I_N - Current Norton

→ For I_N the first thing that needs to be done

* close the gap with a line



$$I_N = \frac{V_1}{10}$$

* then calculate I_N

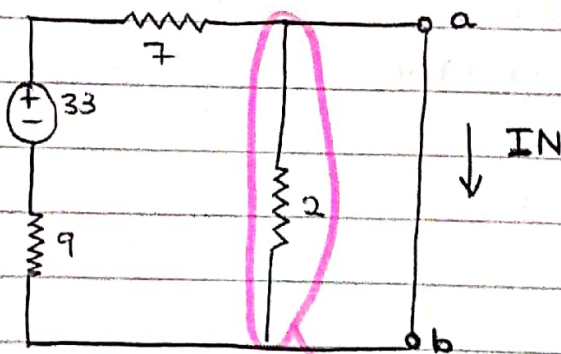
@ V_1 :

$$\frac{V_1 - 33}{16} + \frac{V_1 - 40}{2} + \frac{V_1}{10} = 0$$

$$\frac{V_1}{16} - \frac{33}{16} + \frac{V_1}{2} - \frac{40}{2} + \frac{V_1}{10} = 0$$

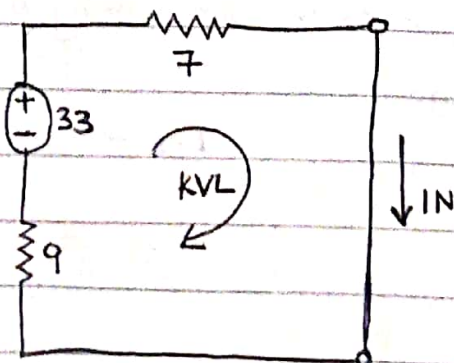
$$\frac{V_1}{16} + \frac{V_1}{2} + \frac{V_1}{10} = \frac{33}{16} + \frac{40}{2}$$

Example 2



* calculate I_N

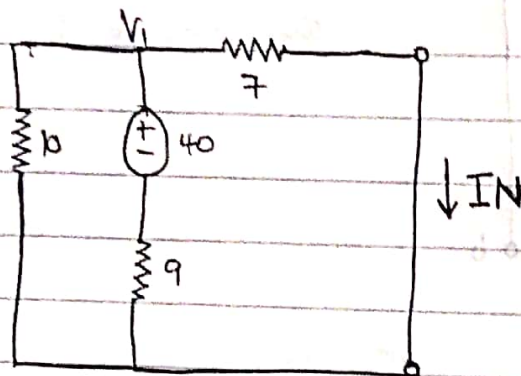
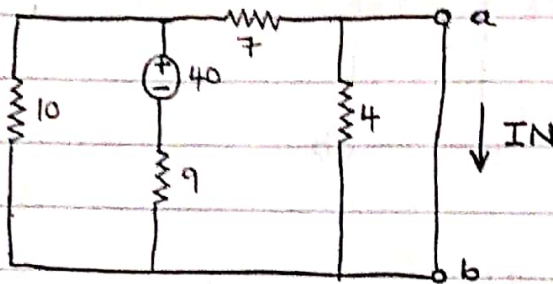
remove b.c. $V=0V$



$$-33 + 16I_N = 0$$

$$I_N = 2.06 \text{ A}$$

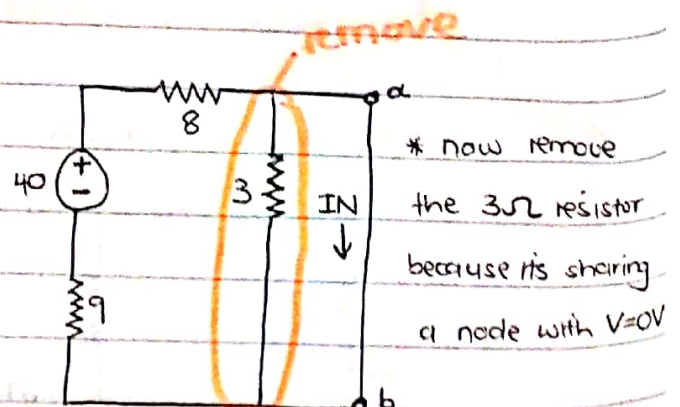
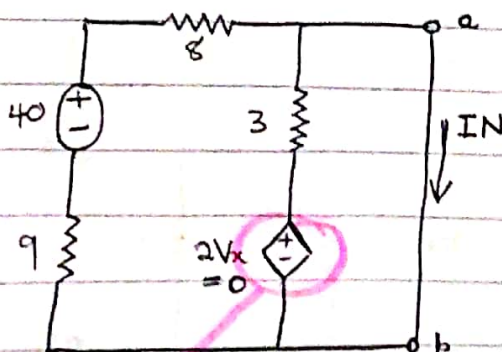
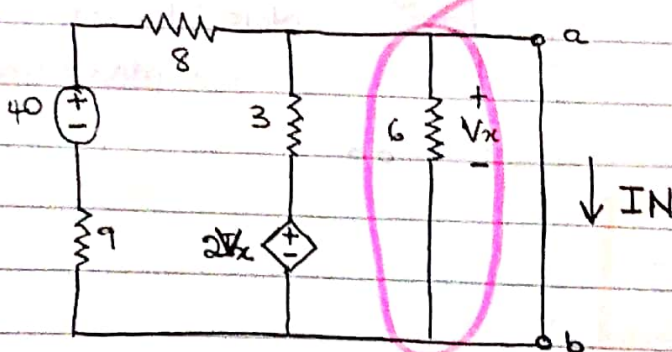
Example 3



Note: we don't remove 10Ω
because it does not share
the same node with the line
with $0V$
$$I_N = \frac{V_1}{7}$$

Do nodal analysis @ V_1

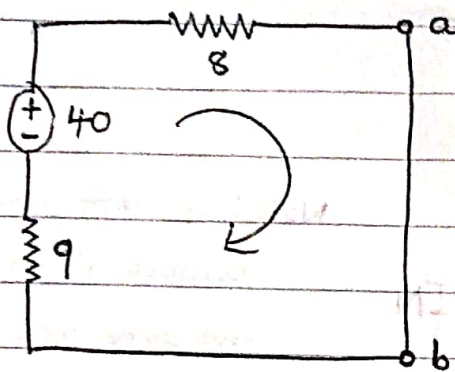
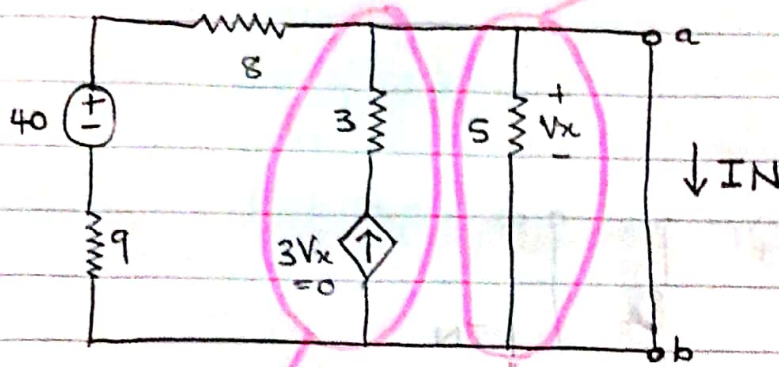
Example 4



* Now remove
the 3Ω resistor
because it's sharing
a node with $V=0V$

We know the voltage value therefore it's now an independent source. $\therefore$ we put a line

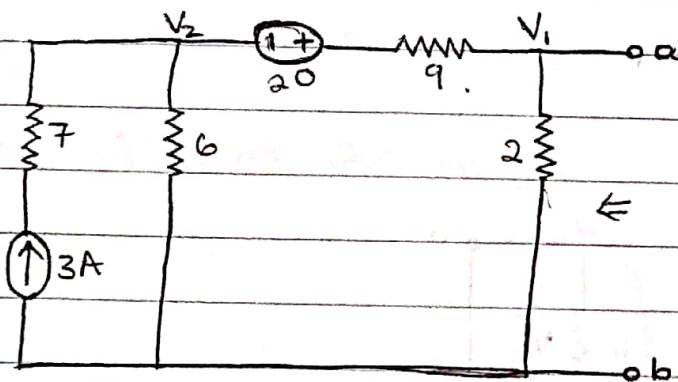
Example 5



then calculate !! SSC!

Maximum Power.

Ex 1

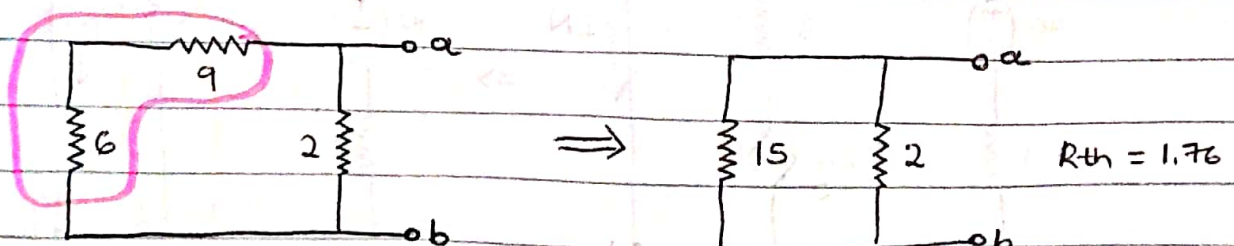


$$P = \frac{V^2}{R} = VI = \frac{W}{E}$$

Note: It's not just **POWER**
but **MAXIMUM POWER**

$$P_{\max} = \frac{V_{th}^2}{4R_{th}}$$

* calculate R_{th} and V_{th}



$$@V_1 : \frac{V_1 - 20 - V_2}{9} + \frac{V_1}{2} = 0$$

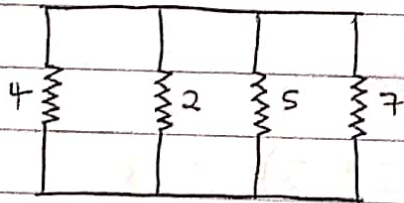
$$@V_2 : \frac{V_2 + 20 - V_1}{9} + \frac{V_2}{6} - 3 = 0$$

$$V_1 = \quad \quad \quad V_2 =$$

$$V_1 = V_{th}$$

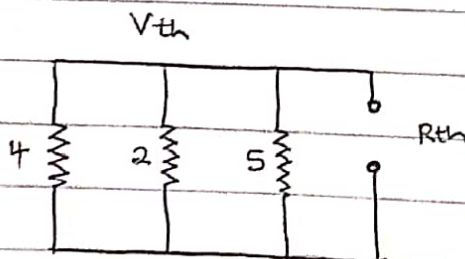
$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$

Example 2

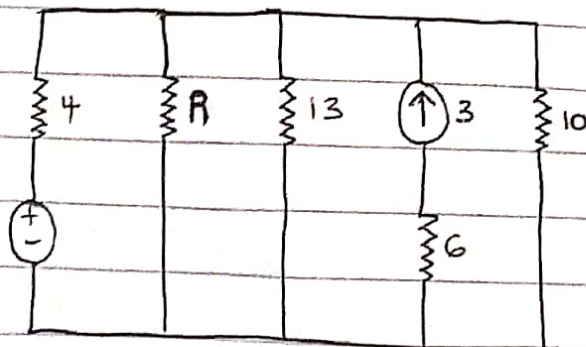


If P_{max} is said to be on the 7Ω resistor
It means $7\Omega = R_{th}$.

* remove the 7Ω resistor and calculate V_{th} .



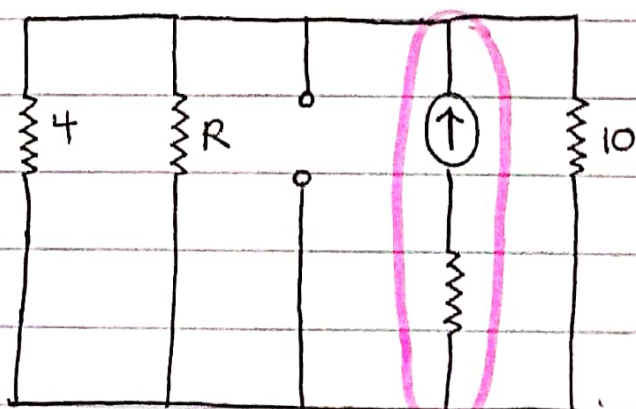
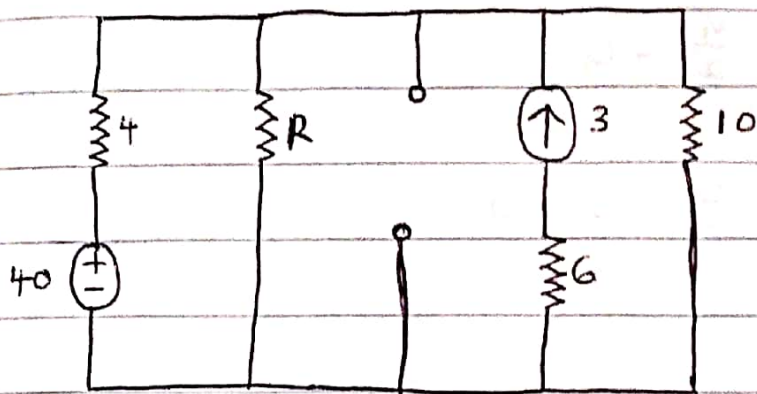
Example 3



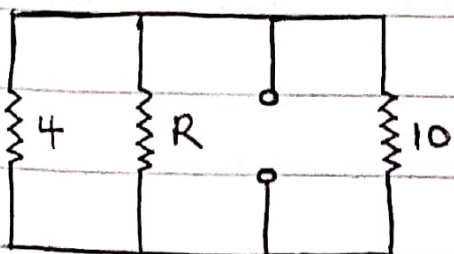
* Maximum power is carried by
 13Ω , Find R ?

$$R_{th} = 13$$

therefore remove the 13Ω



remove



$$\left(\frac{1}{4} + \frac{1}{R} + \frac{1}{10} \right)^{-1} = 13$$

$$R = 6.74 \, \Omega$$